COMPOSITION AND CANCELLATION OF THE COHEN-MACAULAY PROPERTY FOR MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6310-6327 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6329-6349, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohen-Macaulay morphisms of algebraic spaces, etale charts, and flatness; the reduction is retained.

Source reference. Original lemma algebra-lemma-CM-goes-up: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6349–6349. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-CM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6241–6253. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-composition-CM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6349–6349. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of algebraic spaces over S. Assume that the fibres of f, g, and $g \circ f$ are locally Noetherian. Let $x \in |X|$ with images $y \in |Y|$ and $z \in |Z|$.*

1. *If f is Cohen-Macaulay at x and g is Cohen-Macaulay at $f(x)$, then $g \circ f$ is Cohen-Macaulay at x.*
2. *If f and g are Cohen-Macaulay, then $g \circ f$ is Cohen-Macaulay.*
3. *If $g \circ f$ is Cohen-Macaulay at x and f is flat at x, then f is Cohen-Macaulay at x and g is Cohen-Macaulay at $f(x)$.*
4. *If $f \circ g$ is Cohen-Macaulay and f is flat, then f is Cohen-Macaulay and g is Cohen-Macaulay at every point in the image of f.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Working étale locally this follows from the corresponding result for schemes, see More on Morphisms, Lemma more-morphisms-lemma-composition-CM. Alternatively, we can use the equivalence of (a) and (b) in Lemma lemma-CM. Thus we consider the local homomorphism of Noetherian local rings

$$\mathcal{O}_{Y,\overline{y}}/\mathfrak{m}_{\overline{z}}\mathcal{O}_{Y,\overline{y}} \longrightarrow \mathcal{O}_{X,\overline{x}}/\mathfrak{m}_{\overline{z}}\mathcal{O}_{X,\overline{x}}$$

whose fibre is

$$\mathcal{O}_{X,\overline{x}}/\mathfrak{m}_{\overline{y}}\mathcal{O}_{X,\overline{x}}$$

and we use Algebra, Lemma algebra-lemma-CM-goes-up. □